Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	ProjectKita
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8, Decent Work and Economic Growth (supporting: SDG 9)
Project / Solution Name	Payment Follow-up
Name (Leader)	Muhammad Fikri Bin Mohd Nizan
Email & Phone (Leader)	fikrinizan04@gmail.com (+60-19 886 7107)
University Affiliation(s)	Asia Pacific University of Technology and Innovation (APU)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses that sell on credit wait too long to be paid. In Experian's State of Credit 2025 report, Malaysian SMEs waited an average of 64 days for payment, about six days longer than large corporates. The strain shows: in an April 2026 survey by the SME Association of Malaysia, 45% of SMEs feared serious cash flow problems. The law offers little help, as late-payment interest applies only if it is written into the contract.

We interviewed one owner-run micro rental-car business (4 to 5 staff). Its invoices are mainly B2B, plus some B2C when driver services are paid after delivery. The owner tracks them in Excel and talks to customers mainly on WhatsApp. We could not confirm that invoices are forgotten, so we treat that as a hypothesis. We also assume chasing is awkward with key customers or large corporate buyers. A 2017 SME Media Group and INTI survey found many SMEs lacked the voice to demand payment. This is one case, not a survey.

Existing tools do not close the gap. Accounting software mainly sends timer-based email reminders and needs invoices to live inside it. Owners who run on WhatsApp and Excel are left out, and timer-based reminders tend to treat every customer the same.

Why it matters: late cash limits an SME's ability to pay staff, suppliers and rent, which affects job security and growth. This supports SDG 8, especially target 8.3 on micro, small and medium enterprises, and SDG 9.

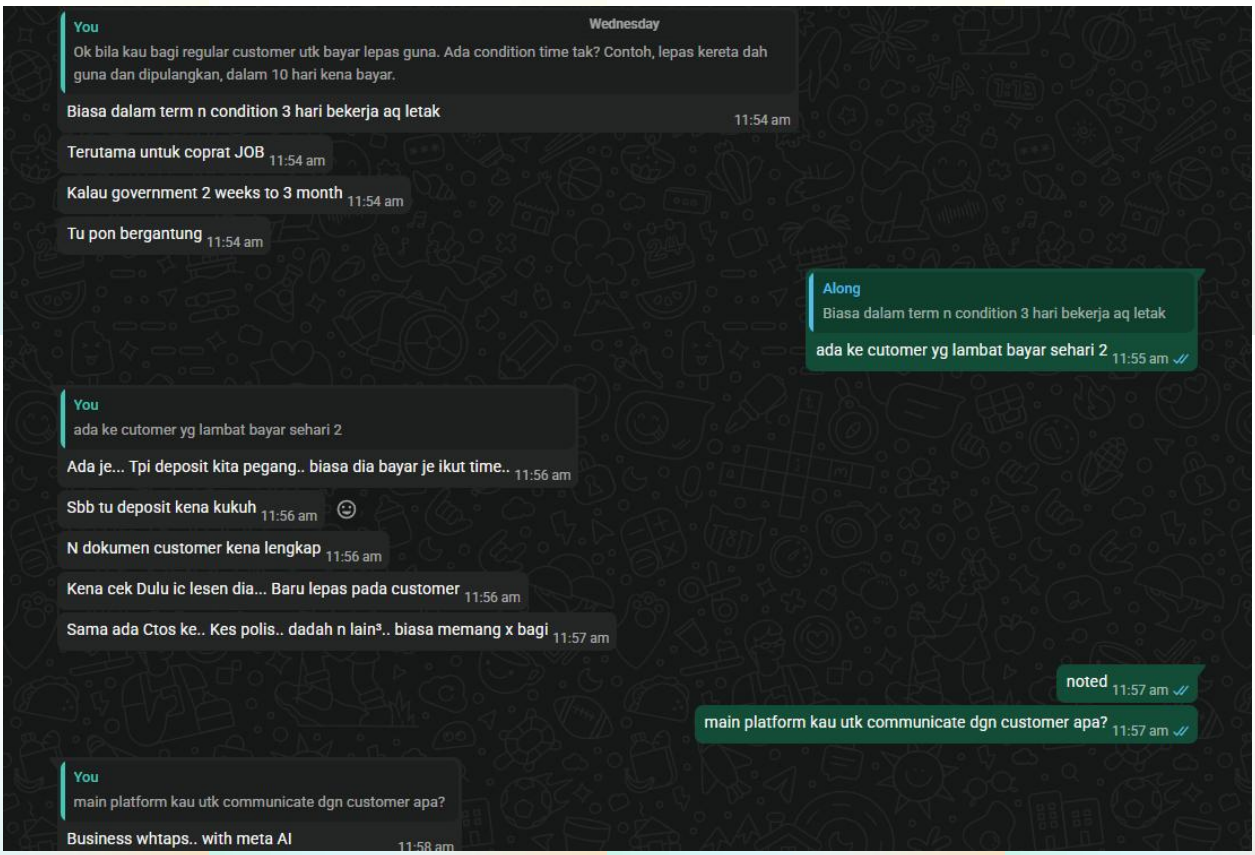
References

- Experian Malaysia, State of Credit 2025: https://www.experian.com.my/2025/05/21/experian-malaysias-state-of-credit-2025-reveals-sme-resilience-but-cash-flow-strains-persist/
- RinggitPlus coverage of Experian: https://ringgitplus.com/en/blog/malaysias-finance-market/payment-gap-widens-between-smes-and-corporates-experian-warns.html
- Coface Asia Payment Survey 2025 (PDF): https://www.coface.hu/content/download/80703/file/Coface%20Asia%20Payment%20Survey%202025.pdf
- Coface Asia Payment Survey 2025 (web): https://www.coface.com/news-economy-and-insights/asia-payment-survey-2025-companies-expect-payment-behaviors-to-worsen-amid-economic-uncertainty
- SME Association of Malaysia (May 2026): https://www.malaysiasme.com.my/sme-malaysia-urges-swift-procurement-reforms-to-ease-sme-cash-flow-crisis/
- The Star, Airwallex SME Outlook 2026: https://www.thestar.com.my/business/business-news/2026/10/01/rising-costs-top-concern-for-malaysian-smes-as-growth-remains-priority
- Allianz Trade, Malaysia: https://www.allianz-trade.com/en_global/economic-research/collection-complexity/malaysia.html
- The SME & INTI (2017): https://sme.asia/tackling-the-ever-present-issue-of-late-payments-to-smes/
- The Sun, WhatsApp overtakes email: https://thesun.my/business/local-business/whatsapp-overtakes-email-as-malaysian-smes-rethink-customer-engagement/
- Malaysia SME, WhatsApp Business Sessions: https://www.malaysiasme.com.my/?p=8012

Appendix for Problem Context



Company Information



Interview Coversation

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Payment Follow-up is a plugin for Orbyte's Work module. Each day it tells an SME owner which customer to chase, why, and what to say. A person approves before anything is sent.

How it works:

1. A daily check, or a Refresh button, scores every overdue invoice: amount still owed × days overdue, after grace days set by customer type.
2. The system gathers the invoice, payments and customer profile, and rules decide who is alerted.
3. AI drafts the reminder in the customer's preferred channel, language (English or Bahasa Malaysia) and tone.
4. Owner or Finance approves, edits, skips or snoozes it.
5. The send is simulated, and every step is saved in an activity log.

Where AI is used: only for writing and explaining. It drafts the reminder, explains the ranking, and writes the owner's weekly summary. Code calculates all amounts, dates and scores, and every number in a draft is checked against the data. If the check or the AI fails, a fixed template is used. AI never sends anything.

What the workshops shaped:

- **Cloudflare and Orbyte:** exploring the Orbyte lab environment shaped our assumed Orbyte data design (clients, projects, users), which we simulate through an adapter layer.
- **Replit:** we use it for deployment and hosting only, because its credit limits could run out mid-development.
- **Runpod:** we serve an open-source language model there, rather than calling a third-party AI API, and send it only the fields it needs.

Stack: FastAPI, PostgreSQL and React. Roles (Owner, Finance, Other staff) are simulated and checked in the backend.

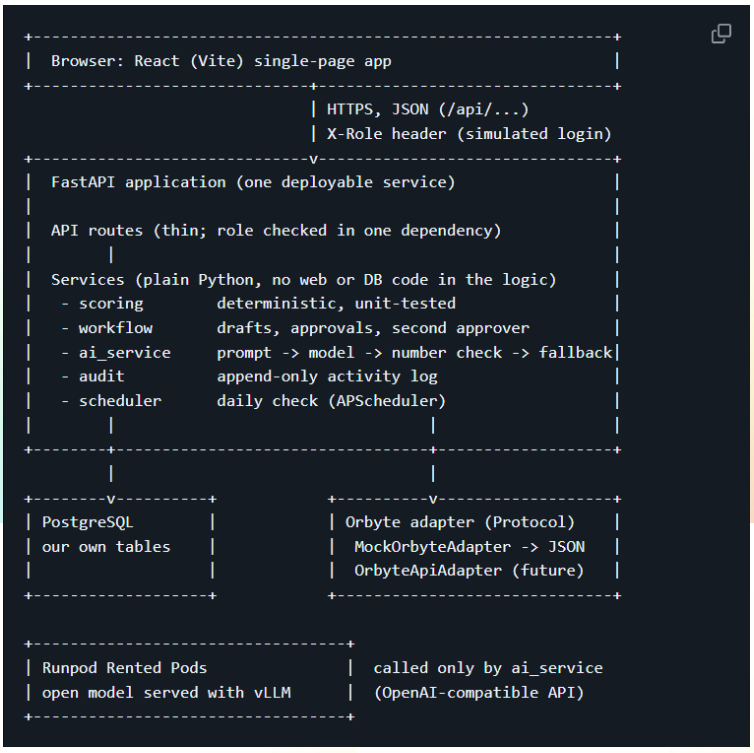
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology
Frontend	React (Vite), Tailwind CSS, shadcn/ui, Recharts
Backend	Python, FastAPI, Pydantic, APScheduler (daily check)
Database	PostgreSQL (SQLAlchemy 2.0), money stored as exact decimals in MYR
AI	Open-source language model served with vLLM on a Runpod GPU Pod, called through the openai Python library
Orbyte simulation	Adapter layer reading synthetic JSON data (swappable for Orbyte's real APIs)
Hosting	Replit (Reserved VM) with its own PostgreSQL
Quality and source control	pytest for the scoring logic, GitHub (main, dev, personal branches)



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: We interviewed an owner-run micro rental-car business to understand how it invoices, tracks payments in Excel and contacts customers on WhatsApp.

Define: One problem: owners lack a consistent way to decide who to chase and how, while staying in control.

Ideate: Our ideas changed after review. Our first priority formula (days late ÷ amount owed) ranked big debts below small ones, so we replaced it with amount owed × days overdue. We also removed a separate drafting role, because it added a second review of the same message.

Prototype: We build on synthetic data with FastAPI, PostgreSQL and React. Orbyte's data is simulated through an adapter layer, so the real platform can replace it later.

Test: Our UAT member will have first-time users complete the main journey unaided, recording time taken and mistakes.

Deployment: The prototype is hosted on Replit with its own PostgreSQL, and code is managed on GitHub with separate main and dev branches. Security controls are already in place: security headers on the frontend, and database row-level security so each company's data stays separate. Role checks are enforced in the backend.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8.3:** support for entrepreneurship and the growth of micro, small and medium-sized enterprises. **SDG 8.2:** higher productivity through technological upgrading and innovation. (Supporting: SDG 9.)
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysian SMEs wait an average of **64 days** to be paid, compared with 58 days for large corporates (Experian Malaysia, *State of Credit 2025*, 21 May 2025). 45% of SMEs feared serious cash flow problems in an April 2026 survey (SME Association of Malaysia, 21 May 2026). Owners currently chase manually through Excel and WhatsApp, with no consistent rule for who to chase first.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary KPI: Owner minutes spent per overdue invoice** (find the invoice, decide whether to chase, write the message). **Secondary KPI:** % of overdue invoices followed up within 3 days of the due date. Both are measured on synthetic data.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	**50% less owner time per overdue invoice** versus a manual baseline the team times on the same synthetic dataset, and **at least 95% of invoices above the alert**

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	**threshold followed up within 3 days.** These are team targets on synthetic data, not real-world results.

Sources

- Experian Malaysia, State of Credit 2025: https://www.experian.com.my/2025/05/21/experian-malaysias-state-of-credit-2025-reveals-sme-resilience-but-cash-flow-strains-persist/
- SME Association of Malaysia (May 2026): https://www.malaysiasme.com.my/sme-malaysia-urges-swift-procurement-reforms-to-ease-sme-cash-flow-crisis/

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.12+	Code (language runtime)	PSF License
FastAPI	Code (framework)	MIT
React 19	Code (framework)	MIT
PostgreSQL	Code (database)	PostgreSQL
SQLAlchemy	Code (library)	MIT
Pydantic	Code (library)	MIT
Vite	Code (build tool)	MIT

TypeScript	Code (language)	Apache 2.0
Tailwind CSS	Code (framework)	MIT
Recharts	Code (library)	MIT
Lucide React	Code (library)	ISC
APScheduler	Code (library)	MIT
OpenAI Python SDK	Code (library)	MIT
Qwen2.5-Coder 1.5B	Model (pre-trained LLM)	Apache 2.0
pytest	Code (library)	MIT
Playwright	Code (library)	Apache 2.0
Docker	Code (platform)	Apache 2.0
Alembic	Code (library)	MIT
psycopg2	Code (library)	LGPL

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



(24/10/2026)